



SCRAPIFY ECOTECH PVT. LTD.

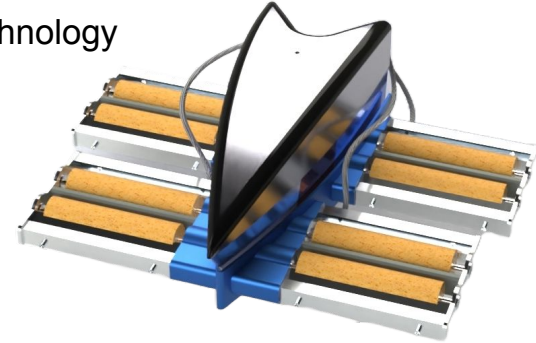
ECOFLOATER

AI-Powered Smart, Sustainable, Scalable Water Drone Technology

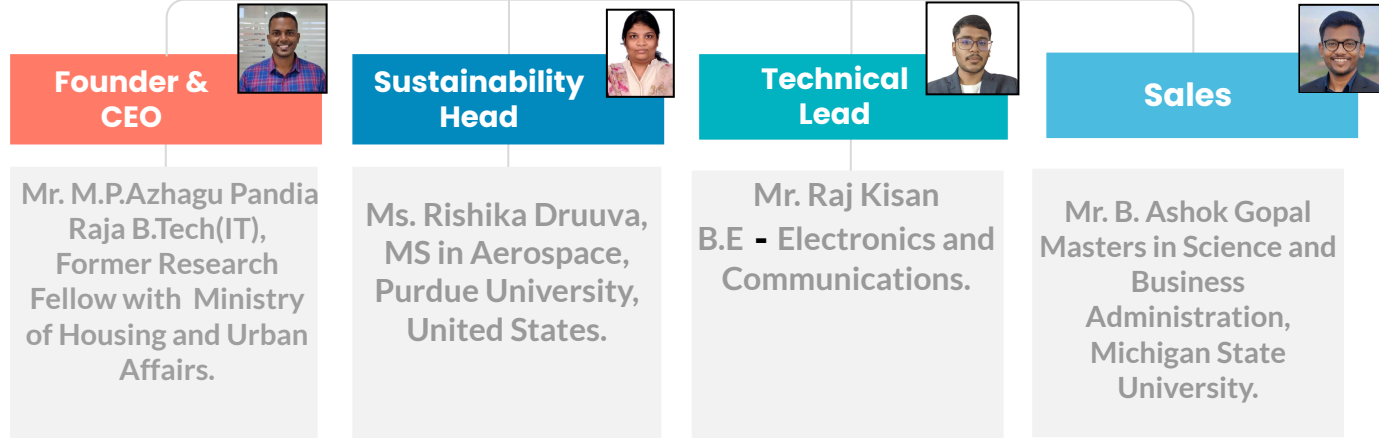


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OUR TEAM



Founder's Profile

Azhagu Pandia Raja, a former Research Fellow for the India Smart City Mission under the Ministry of Housing and Urban Affairs, Government of India, contributed to projects such as IoT-enabled Integrated Command and Control Centers and the India Waste Exchange. He became India's first City Innovation Officer for the Greater Chennai Corporation, where he led the IT team in combating the spread of COVID-19. During this time, he also designed and implemented various digital tools at the city level for monitoring and controlling the pandemic.

Link :
<https://www.thenewsminute.com/article/meet-india-s-first-innovation-officer-greater-chennai-corporation-s-azhagu-pandia-raja>



ACHIEVEMENTS

Monetary Achievements :



Non- Monetary Achievements :



Niti Aayog Lifeathon

Top 75 Global idea in 'Thinking for the Planet: 75 ideas to promote LiFE'



Denmark Tech Fest

Represented Indian contingent at Denmark Tech Fest Oct, 2024.



TIE50 Winner

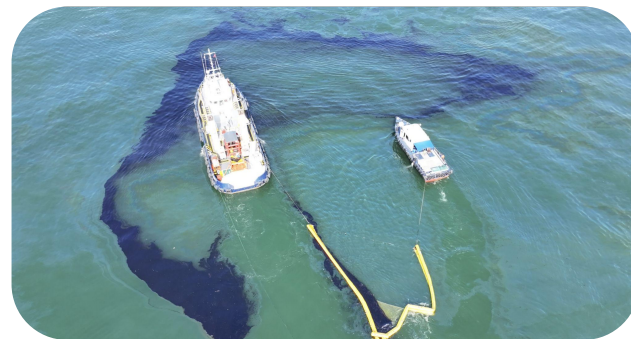
One among the Top TiE'50 Winners, TiE con 2025.



PROBLEMS ADDRESSED

Oil and Chemical Spill Containment & Recovery

- Oil and chemical spills spread quickly, contaminating water and harming aquatic ecosystems
- Toxic substances pose serious health risks to humans, wildlife, and fisheries
- Traditional cleanup methods are slow, costly, and require skilled labor
- Delays in detection worsen environmental and economic damage
- Lack of automated, responsive systems for localized containment and cleanup
- Limited real-time monitoring and reporting of spill spread and impact





PROBLEMS ADDRESSED

Floating debris and invasive weeds cleanup

Floating debris and invasive weeds degrade water quality, demanding labor-intensive cleanup.



Limited Bathymetric Mapping

Hydrographic surveys are rare, expensive, and require heavy equipment or skilled manpower.



Manual Aquaculture Operations

Fish feeding is mostly manual, prone to over / underfeeding, and lacks timing precision and nutrient tracking.



Port Surveillance Gaps

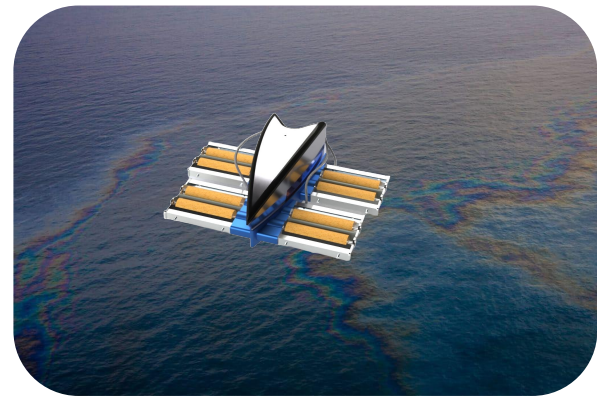
Static cameras miss blind spots; unauthorized intrusions and accidents go unnoticed in real time.





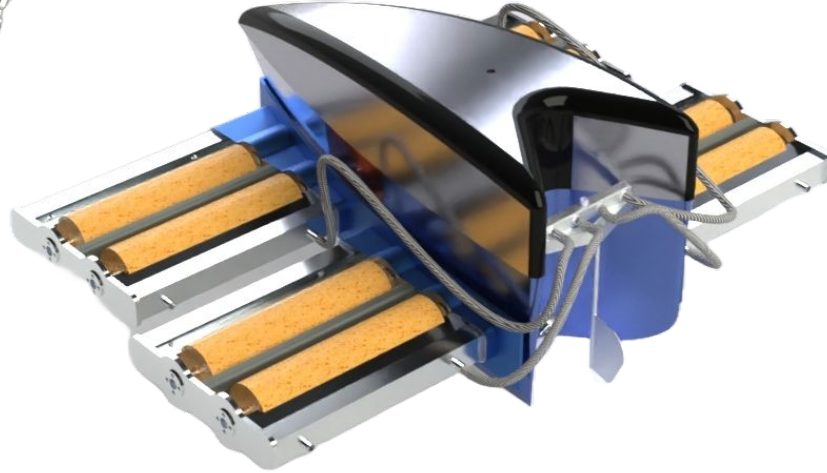
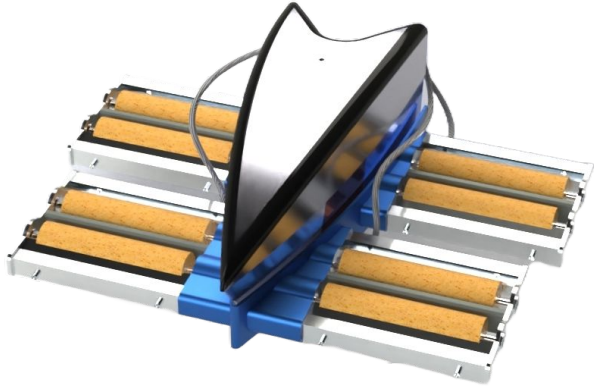
Oil skimmer

- Equip Eco Floater with a **modular oil and chemical skimming unit** to capture surface contaminants efficiently
- Use **hydrophobic absorbent rollers or collection pads** to separate oil/chemicals from water without disturbing aquatic life
- Enable **real-time spill detection** using onboard sensors (e.g., hydrocarbon, pH, turbidity) and AI-based visual monitoring
- Support **automated deployment** to affected zones using GPS and remote control or autonomous navigation
- Log spill data (location, spread, volume) for **regulatory reporting and rapid response coordination**
- Reduce human exposure by enabling **contactless cleanup operations** in hazardous spill zones
- Design system for **easy replacement of skimmer pads and waste containers**, ensuring continuous operation in the field





OIL SKIMMER DESIGN





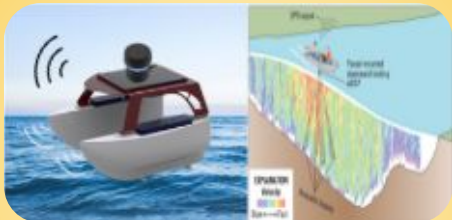
OUR SOLUTION

Floating Waste Collection



Eco Floater uses AI based autonomous waste collection mechanisms to remove floating debris and weeds efficiently.

Bathymetric Survey & Mapping



Integrated sonar sensors enable real-time, lightweight, and low-cost depth profiling during routine operations.



ECO-FLOATER

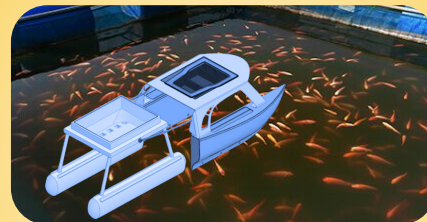
Clean, Map, Monitor, Survey, Fish Feed

Port Surveillance and Security



Equipped with mobile vision systems, Eco Floater eliminates blind spots for continuous port and perimeter monitoring.

Automated Fish Feeding



Automated feeders with sensor-based schedules ensure precise, optimized feeding with nutrient and time tracking.

Eco Floater Oil & Chemical Spill Skimmer

- **Autonomous Skimming:** Enables unmanned, GPS-guided response for oil and chemical spill containment
- **Modular Integration:** Attachable skimmer module fits seamlessly on the Eco Floater platform
- **AI-Based Visual Detection:** Uses YOLO model for real-time visual identification of oil/chemical spread on water surfaces
- **Smart Sensor Suite:** Integrated with hydrocarbon, turbidity, and pH sensors for spill confirmation and monitoring
- **Eco-Friendly Recovery:** Utilizes hydrophobic rollers or absorbent pads to clean spills without harming aquatic ecosystems
- **Dual-Mode Operation:** Supports autonomous and manual control for targeted spill recovery
- **Compact & Rapid Deployment:** Ideal for quick response in ports, lakes, and industrial zones—no heavy equipment needed
- **Automated Data Logging:** Tracks location, extent, and volume of spills for environmental reporting and regulatory compliance
- **Minimizes Human Risk:** Reduces exposure to toxic or flammable substances during cleanup operations

S.No	Operations	USP
1	Waste Collection	An autonomous, modular platform with AI-powered waste detection and remote/manual control, designed for efficient water surface cleaning across varied environments.
2	Water Quality Monitoring	Provides continuous, geo-tagged multi-parameter sensing with real-time alerts and AI-driven pollution prediction, enabling proactive water health management.
3	Port Surveillance	Delivers dynamic 360° mobile surveillance with autonomous patrolling and instant intrusion or hazard alerts, eliminating blind spots of static camera systems.
4	Aquaculture Support	Optimizes fish feeding and health with automated nutrient delivery, water quality tracking, and remote monitoring tailored to aquaculture conditions.
5	Bathymetric Mapping	Offers lightweight, real-time sonar-based depth mapping integrated into routine operations, eliminating the need for costly survey vessels and crews.
6	Data & Intelligence Layer	Integrates all environmental, operational, and visual data into a unified edge-cloud system with AI analytics and API support for informed, data-driven decision-making.



TARGET CUSTOMERS



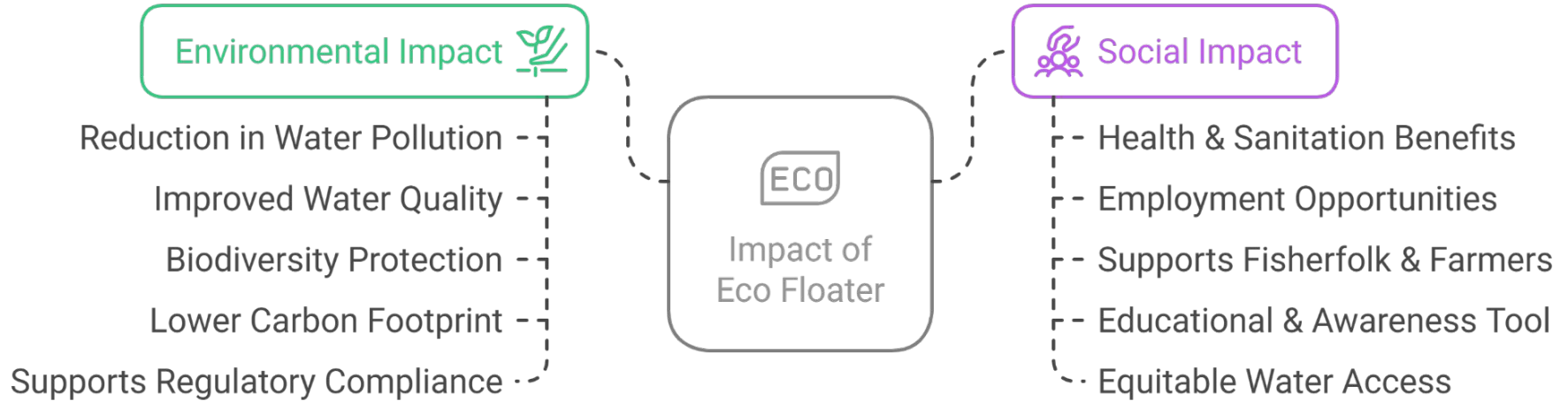


BUSINESS MODEL

Beneficiaires	The beneficiaries of these addressed problem are Municipal bodies, port authorities, fisheries departments, aquaculture farmers, environmental agencies, industrial operators, research institutions, and local communities dependent on clean and healthy water ecosystems.			
Target Market Size	Market Size	Target Market Size (INR Crores)	Addressable Market Size(%)	Estimated Addressable Market value(INR crores)
	India	₹2,76,000+	25%	₹1,40,000+
	Tamil Nadu	₹14,087+	30%	₹7,145+
	Chennai	₹1,575+	20%	₹818.65+
Customer Segments / Subsegments	Type of Business: Business-to-Business (B2B) and Business-to-Government (B2G) model, offering multipurpose autonomous water drones for cleaning, monitoring, surveillance, and aquaculture support to industries, government bodies, and environmental agencies.		Target Customer: Municipal bodies, port authorities, fisheries, environmental agencies, industrial facilities, and research institutions.	
Impacts & Outcomes	Eco Floater enables real-time waterbody management through waste collection, water monitoring, mapping, surveillance, and smart aquaculture using a single modular platform.			



IMPACT





MILESTONES WITH TIMELINE

Phase of the project	Milestones planned	Description
Phase 1 Advanced System Design & Strategic Planning (1 to 3rd Month)	1. Multipurpose System Requirements Finalization	→ Define functional specs across all modules: waste, water quality, mapping, surveillance, and aquaculture.
	2. Integrated Architecture & Component Selection	→ Finalize modular hardware design, AI architecture, sensor stack, and thruster configurations suitable for multipurpose operations.
	3. CAD Modeling and Structural Simulation	→ Design full-scale body, modular conveyor, and bridge with simulations for buoyancy, load distribution, and sensor payload accommodation.
Phase 2 Full-Scale Prototype Fabrication & Core AI Integration (4 to 6th Month)	1. Hardware Assembly with Modular Attachments	→ Fabricate hulls, bridges, and mount modules for conveyor, sensors and aquaculture unit with actuator interfaces.
	2. Sensor, Electronics & Communication Integration	→ Install and wire up GPS, IMU, sonar, water quality sensors, cameras, and wireless comms via Raspberry Pi & Jetson Nano.
	3. Core AI/ML Module Implementation	→ Develop vision model for waste detection, water contamination detection, and fish-feeding logic with real-time edge inference.



MAILESTONES WITH TIMELINE

Phase of the project	Milestones planned	Deliverables
Phase 3 Autonomous Navigation, Control Systems & In-House Testing (7 to 9 th Month)	1. AI-Based Navigation & Module Coordination	→Implement sensor fusion for autonomous navigation with dynamic task-switching across modules (e.g., cleaning vs mapping).
	2. Simulated Environment Testing	→ Run controlled environment tests for path planning, module switching, and data collection accuracy.
	3. Dashboard + Manual Override Interface	→Develop unified dashboard for live telemetry, remote control, and switching between operational modes.
Phase 4 Field Trials, Optimization & Pre-Deployment Readiness (10 to 12 th Month)	1.Multi-Use Field Trials	→ Pilot real-world trials across ports, lakes, and fish farms covering all functional modules with partner agencies.
	2. System Calibration & Feedback Loop	→ Fine-tune AI accuracy, control responsiveness, and modular transitions based on live feedback.
	3. Commercial Launch & Sales Enablement	→ Execute go-to-market plan, onboard B2B/B2G clients, and initiate partnerships across sectors (municipal, industrial, aquaculture, etc.).



GROWTH PLAN

S.NO	YEAR (FY)	UNIT SOLD	REVENUE (IN CRS.)	NET PROFIT (IN CRS.)
1	2025-26	15	₹7.5	-₹1.0
2	2026-27	60	₹30	₹ 2.0
3	2027-28	150	₹75	₹ 12.0
4	2028-29	300	₹150	₹ 28.0
5	2029-30	500	₹250.0	₹ 50.0



Sales Target

Aim to sell over 500 units by year 2029



Contract Acquisition

Secure contracts in 5+ states



Revenue Goal

Achieve ₹500 Cr in revenue over 5 years



Profitability Milestone

Break-even by Year 2; maintain 20% EBITDA from Year 3



Global Expansion

Post - Financial Yr. 2028 - 29 international market entry



India Market Share

20-30% target by Year 5



India SAM

₹3,700 Cr. market size